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Method for manufacturing gate including boy part and protruding part and gate structure

Abstract

A method for manufacturing a semiconductor structure and the semiconductor structure are provided. The method for manufacturing a semiconductor structure includes: providing an activated region; forming an initial gate located on the activated region; forming a first mask layer on a top surface of the initial gate, in which a first opening penetrating the first mask layer is provided in the first mask layer, and the first opening at least has opposite two sides extending along a first direction; forming sidewall layers located at least on sidewalls of both sides of the first opening extending in the first direction; removing the first mask layer; patterning the initial gate with the sidewall layers on both sides of the first opening as a mask to form gates.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of International Application No. PCT/CN2022/093394, filed on May 17, 2022, which claims priority to Chinese Patent Application No. 202210434946.3, filed on Apr. 24, 2022. The disclosures of International Application No. PCT/CN2022/093394 and Chinese Patent Application No. 202210434946.3 are incorporated herein by reference in their entireties.

BACKGROUND

- (1) Memory is a memory component used to store programs and various data information. Random access memory (RAM) used in general computer systems may be divided into dynamic random access memory (DRAM) and static random-access memory (SRAM). Dynamic random access memory is a semiconductor memory device commonly used in computers, consisting of many repetitive memory cells.
- (2) A memory cell typically includes a capacitor and a transistor, different devices, such as sense amplifier, are formed by electrical connections between the different transistors.
- (3) However, with the continuous shrinking of the size, the gate is becoming smaller and smaller. In the case of shrinking of the gate, how to achieve the effective connection between the gate and the contact structure has become an urgent problem to be solved.

SUMMARY

- (4) Embodiments of the disclosure relate to the field of semiconductor, in particular to a method for manufacturing a semiconductor structure and the semiconductor structure.
- (5) According to some embodiments of the disclosure, in one aspect, the embodiments of the disclosure provide a method for manufacturing a semiconductor structure. The method includes: providing an activated region; forming an initial gate located on the activated region; forming a first mask layer on a top surface of the initial gate, in which a first opening penetrating the first mask layer is provided in the first mask layer, and the first opening at least has opposite two sides extending along a first direction; forming sidewall layers located at least on sidewalls of the two sides of the first opening extending in the first direction; removing the first mask layer; patterning the initial gate with the sidewall layers at the two sides of the first opening as a mask to form gates.
- (6) According to some embodiments of the disclosure, in another aspect, the embodiments of the disclosure further provide a semiconductor structure. The semiconductor structure includes an activated region; a gate located on the activated region, in which the gate includes a body part extending in the first direction and a protruding part located on one side of the body part, the protruding part is connected with a sidewall of the body part extending along the first direction, and the protruding part and the body part are in an integrated structure; a contact structure located on a surface of the protruding part and electrically connected with the gate.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) One or more embodiments are exemplary illustrated by the figures in the corresponding drawings, which are not to be limiting to the embodiments, and the figures in the drawings are not to be scale limiting unless otherwise stated. In order to more clearly illustrate the embodiments of the disclosure or the technical solution in the conventional technique, a brief description of the drawings required to be used in the embodiments will be provided below. Apparently, the drawings of the following description are merely some embodiments of the disclosure. For a person of ordinary skill in the art, other drawings can be obtained based on these drawings without creative work.

(2) FIGS. **1-14** are structural schematic diagrams corresponding to various operations in a method for manufacturing a semiconductor structure provided by an embodiment of the disclosure;

(3) FIGS. **15-18** are structural schematic diagrams corresponding to various operations in another method for manufacturing a semiconductor structure provided by an embodiment of the disclosure;

(4) FIGS. **19-24** are structural schematic diagrams corresponding to various operations in a method for manufacturing a semiconductor structure provided by another embodiment of the disclosure;

(5) FIGS. **25-28** are structural schematic diagrams corresponding to various operations in another method for manufacturing a semiconductor structure provided by another embodiment of the disclosure;

(6) FIG. **29** is a schematic diagram of a semiconductor structure provided by an embodiment of the disclosure;

(7) FIG. **30** is a schematic diagram of another semiconductor structure provided by an embodiment of the present disclosure.

DETAILED DESCRIPTION

(8) The embodiments of the disclosure provide a method for manufacturing a semiconductor structure. Gates are formed by providing an activated region; forming an initial gate located on the activated region; forming a first mask layer on a top surface of the initial gate, in which a first opening penetrating the first mask layer is provided in the first mask layer, and the first opening at least has opposite two sides extending along a first direction; forming sidewall layers located at least on sidewalls of both sides of the first opening extending in the first direction; removing the first mask layer; patterning the initial gate with the sidewall layers on two sides of the first opening as a mask. Two gates can be formed in the same step by using the sidewall layers on both sides of the first opening as a mask, thereby providing a structural basis for subsequent forming transistors and an amplifier.

(9) The embodiments of the present disclosure will be described in detail below with reference to the accompanying drawings. However, those skilled in the art should understand that in various embodiments of the present disclosure, numerous technical details are proposed for a reader to better understand the present disclosure. But even without these technical details and various variations and modifications based on the following embodiments, the technical solutions claimed in the present disclosure may be implemented.

(10) Referring to FIGS. **1** and **2**, FIG. **1** is a top view of a semiconductor structure provided by the embodiment of the present disclosure, and FIG. **2** is a sectional view along the AA direction in FIG. **1** provided by the embodiment of the present disclosure.

(11) Specifically, an activated region **100** is provided; an initial gate **110** is formed on the activated region **100**.

(12) The activated region **100** may include a substrate **101**, an active area **102** located on the substrate **101** and isolation structures **103** located on opposite two sides of the active area **102**. The activated region **100** may be patterned to form the active areas **102** spaced apart away each other, and then the material corresponding to the isolation structure **103** is deposited to form the isolation structure **103**.

(13) In some embodiments, the material of the substrate **101** may be silicon, germanium, silicon germanium or the like, and the material of the substrate **101** may also be doped. Taken as an example that the material of the substrate **101** is silicon, the substrate **101** is doped with a trace of trivalent elements such as boron, indium, gallium or aluminum, thereby forming a P-type base; Similarly, the substrate **101** is doped with a trace of pentavalent elements, such as phosphorus, antimony, arsenic, or the like, thereby forming an N-type base. The selection of doping elements of the substrate **101** can be considered according to actual requirements and product properties, etc., and the present disclosure does not limit the material of the substrate **101** and the doping elements.

(14) In some embodiments, the isolation structure **103** may include a first isolation structure **104**, a second isolation structure **105** and a third isolation structure **106**. The material of the first isolation structure **104** may be the same as that of the third isolation structure **106**, both are silicon oxide, and the material of the second isolation structure **105** may be silicon nitride, silicon oxynitride, or the like. An ONO structure, i.e. an oxide-nitride-oxide structure, can be formed by forming the first isolation structure **104**, the second isolation structure **105** and the third isolation structure **106**. By forming the ONO structure, a higher relative dielectric constant, a high breakdown electric field, a low leakage characteristic, and the like can be provided.

(15) In some embodiments, the initial gate **110** may include a first initial conductive layer **111**, a second initial conductive layer **112** and an initial protective layer **113** which are stacked, and the first initial conductive layer **111** is located on the top surface of the active areas **102**. The material of the first initial conductive layer **111** may be a semiconductor material such as polysilicon, the material of the second initial conductive layer **112** may be a metal material such as tungsten, and the material of the initial protective layer **113** may be an insulating material such as silicon nitride.

(16) Direct transmission of an electrical signal from monocrystalline silicon to tungsten can be avoided by providing the first initial conductive layer **111**, thereby reducing the possibility of electrical signal distortion. The transmission speed of an electrical signal of the semiconductor structure can be improved by providing the second initial conductive layer **112** as a metallic material, thereby reducing the response time of semiconductor structure. By providing the first initial conductive layer **111**, it can be avoided that the metal of the second initial conductive layer **112** occurs ion diffusion and thus the active area **102** is affected, so that the reliability of the semiconductor structure is improved. By providing the initial protective layer **113**, the stress effect on the first initial conductive layer **111** and the second initial conductive layer **112** can be reduced during the manufacturing of the semiconductor structure, thereby protecting the first initial conductive layer **111** and the second initial conductive layer **112**.

(17) In some embodiments, before the initial gate **110** is formed, the method further includes forming a gate dielectric layer **120**, by which the initial gate **110** and the active areas **102** can be isolated, thereby avoiding direct contact between the gate **110** and the active areas **102**.

(18) In some embodiments, after the initial gate **110** is formed, the method further includes forming an initial intermediate layer **130** and an initial isolation layer **140** on the surface of the initial gate **110**. The initial intermediate layer **130** is located between the initial isolation layer **140** and the initial gate **110**. The material of the initial intermediate layer **130** may be carbon or carbon-containing organic, and the material of the initial isolation layer **140** may be silicon oxynitride.

(19) Forming the initial intermediate layer **130** on the surface of the initial gate **110** can improve the uniformity of the gate subsequently formed, so that the accuracy of subsequent patterning can be improved. However, in some embodiments, the material of the initial intermediate layer **130** is relatively soft, the formed initial intermediate layer **130** has a poor morphology. An initial isolation layer **140** may be formed on the surface of the initial intermediate layer **130**, the material of the initial isolation layer **140** is relatively hard and thus the morphology of the initial intermediate layer **130** can be improved by limiting the position of the initial intermediate layer **130**. The gate can be prevented from contacting with an etching reagent for removing the first mask layer in the subsequent process of removing the first mask layer by forming the initial isolation layer **140**, so

that the gate electrode can be protected and the accuracy of patterning can be improved.

(20) In some embodiments, after the initial isolation layer **140** is formed, the method further includes forming a first initial mask layer **150** which provides a process basis for subsequent formation of the first mask layer.

(21) In some embodiments, after the first initial mask layer **150** is formed, the method further includes forming a first photoresist layer **160**. The first photoresist layer **160** has at least a first pattern that exposes the surface of the first initial mask layer **150**.

(22) The material of the first photoresist layer **160** includes photoresist. Photoresists are classified into positive and negative types according to the images they form. In the photoresist process, after the coating is exposed and developed, the exposed part is dissolved and the unexposed part remains, then the coating material is a positive photoresist. If the exposed part is retained and the unexposed part is dissolved, then the coating material is a negative photoresist.

(23) Embodiments of the present disclosure take a negative photoresist as an example. Trenches are formed on the first photoresist layer **160** by retaining the exposed part.

(24) Referring to FIGS. **3** and **4**, FIG. **3** and FIG. **4** are operations performed on the basis of FIG. **1** and FIG. **2**, and FIG. **4** is a sectional view of FIG. **3** in the AA direction.

(25) The first mask layer **151** is formed on the top surface of the initial gate **110**. A first opening **170** penetrating the first mask layer **151** is provided in the first mask layer **151**, and the first opening **170** at least has opposite two sides extending along the first direction X.

(26) In some embodiments, the first initial mask layer **150** (referring to FIG. **2**) may be patterned with the material of the first photoresist layer **160** (referring to FIG. **2**) as a mask, and the remaining part of the first initial mask layer **150** (referring to FIG. **2**) serves as the first mask layer **151**.

(27) After the first mask layer **151** is formed, the photoresist layer **160** is removed (referring to FIG. **2**).

(28) In some embodiments, the first mask layer **151** further includes second openings **171**. The second openings **171** are located on opposite two sides of the first opening **170** along the second direction Y and in communication with the first opening **170**. The first opening **170** and the second openings **171** provide a process basis for the subsequent formation of the sidewall layer and further provide a process basis for the subsequent formation of the gate.

(29) Referring to FIGS. **5** to **12**, FIGS. **5** to **12** are operations performed on the basis of FIGS. **3** and **4**. FIG. **6** is a sectional view of FIG. **5** along the direction of the dashed line; FIG. **8** is a sectional view of FIG. **7** along the direction of the dashed line; FIG. **10** is a sectional view of FIG. **9** along the direction of the dashed line; FIG. **12** is a sectional view of FIG. **11** along the direction of the dashed line.

(30) Sidewall layers **180** are formed, and the sidewall layers are at least located on sidewalls of both sides of the first opening **170** extending in the first direction X.

(31) In some embodiments, the first mask layer **151** further includes second openings **171**, and forming the sidewall layers **180** includes forming the sidewall layers **180** located on the sidewalls of the first opening **170** and the sidewall layers **180** located on the sidewalls of the second openings **171** in the same step.

(32) The sidewall layers **180** on the sidewalls of the second openings **171** provide a process basis for the subsequent formation of the protruding part of the gate, thereby providing a larger contact area for the subsequently formed contact structure to reduce the contact resistance between the contact structure and the gate, and thereby improving the performance of the semiconductor structure.

(33) In some embodiments, in the operation of forming the sidewall layers **180**, a sidewall layer **180** that fills up the second opening **171** is formed. By filling up the second opening **171**, there is no gap in the protruding part of the gate subsequently formed and the contact area between the protruding part and the contact structure subsequently formed is increased.

(34) In some embodiments, in the second direction, the length of the second opening **171** is larger than the width of the sidewall layer **180** on the sidewall of the first opening **170**. It could be understood that, if the length of the second opening **171** is less than the width of the sidewall layer **180**, in the process of forming the sidewall layer **180** on the sidewall of the second opening **171**, since the length of the second opening **171** is small, the material of the sidewall layer **180** cannot be fully accommodated along the length of the second opening **171**, and part of the material of the sidewall layer **180** may accumulate in a direction away from the activated region **100** during deposition, which results in that the surface of the sidewall layer **180** has different heights, and even part of the material of the sidewall layer **180** covers some unnecessary areas, so that the pattern forming the gate subsequently is inaccurate. By providing the length of the second opening **171** larger than the width of the sidewall layer **180** on the sidewall of the first opening **170**, the formed sidewall layer **180** can have a better morphology.

(35) In some embodiments, the material of the sidewall layers **180** includes at least one of silica or silicon nitride. Taking the material of the sidewall layer **180** as silicon dioxide as an example, the sidewall layer **180** is formed on the sidewalls of the first opening **170** and the second opening **171** by depositing silica. Since silica is softer, the formed sidewall layer **180** has a good morphology.

(36) Specifically, referring to FIGS. **5** and **6**, an initial sidewall layer **181** is formed, which is located on the sidewalls of the first opening **170** and a bottom of the first opening **170**, and the top surface of the initial sidewall layer **181** in the first opening **170** is lower than the top surface of the first mask layer **151**.

(37) The initial sidewall layer **181** may be formed by depositing silica on the sidewalls of the first opening **170**, and the formed initial sidewall layer **181** is lower than the top surface of the first mask layer **151**, which may improve the accuracy of the subsequent formation of the sidewall layer, and thus improve the pattern accuracy of the gate subsequent formed, and thereby improving the reliability of the semiconductor structure.

(38) In some embodiments, in the operation of forming the initial sidewall layer **181**, an initial sidewall layer **181** is formed on the top surface of the first mask layer **151**.

(39) In some embodiments, the first mask layer **151** further includes second openings **171**, so that the initial sidewall layer **181** is formed further on the sidewalls of the second openings **171**. The formation of the initial sidewall layer **181** on the sidewalls of the second openings **171** provides a process basis for the subsequent formation of the gate having protruding parts, thereby providing more contact area for the contact structure subsequently formed.

(40) In some embodiments, the opening width of the second openings **171** in the second direction Y is larger, so that the filling material of the initial sidewall layer **181** does not fill up the second opening **171** when the initial sidewall layer **181** is formed, and a space is formed in the second opening **171**. In other embodiments, the opening width of the second openings **171** in the second direction Y is smaller, so that the filling material of the initial sidewall layer **181** fills up the second openings **171** when the initial sidewall layer **181** is formed.

(41) Referring to FIGS. **7** and **8**, the redundant initial sidewall layer **181** and the first mask layer **151** are etched away, so that the remaining initial sidewall layer **181** is only the initial sidewall layer **181** located on the sidewalls of the first mask layer **151**, thereby providing a process basis for the subsequent formation of the sidewall layers spaced apart away each other.

(42) The initial sidewall layer **181** located at the part of the bottom of the first opening **171** and the top surface of the first mask layer **151** is removed at the same time by maskless etching process, that is, the initial sidewall layer **181** is formed on the sidewalls of the first opening **171** by a sidewall process (Spacer process), so that the initial sidewall layer **181** with a more accurate pattern can be formed, and the initial sidewall layer **181** formed on the sidewalls of the first opening **171** also defines an opening having a shape similar to that of the first opening **171**.

(43) Referring to FIGS. **9** to **12**, the initial sidewall layer **181** located at part of the bottom of the first opening **170** (referring to FIG. **5**) is removed, and the initial sidewall layer **181** located on

opposite sidewalls of the first opening **170** (referring to FIG. 5) in the first direction X is removed to form the initial sidewall layers spaced in a direction perpendicular to the first direction X, and the remaining initial sidewall layer serves as the sidewall layers.

(44) Referring to FIGS. 9 and 10, the process of removing the initial sidewall layer **181** located on opposite sidewalls of the first opening **170** in the first direction X includes: forming a second mask layer **190** after the first mask layer **151** (referring to FIG. 6) is removed, in which the second mask layer **190** exposes the initial sidewall layer **181** on the opposite sidewalls of the first opening **170** along the first direction X.

(45) In some embodiments, the second mask layer **190** also exposes a part of the surface of the initial isolation layer **140**. It can be ensured that the initial sidewall layer **181** on the opposite sidewalls of the first opening **170** in the first direction X can be completely exposed by the second mask layer **190** exposing a part of the surface of the initial isolation layer **140**, thereby ensuring that the initial sidewall layer **181** on the opposite sidewalls of the first opening **170** in the first direction X can be completely removed in a subsequent etching process. In other embodiments, the second mask layer may not expose the second mask layer inside the first opening, and the present disclosure does not limit the relative positions of the second mask layer **190** and the initial isolation layer **140**, only requiring that the initial sidewall layer **181** on the opposite sidewalls of the first opening **170** in the first direction X can be completely exposed.

(46) In some embodiments, the first mask layer **151** (referring to FIG. 6) also includes second openings **171** (referring to FIG. 5), and the second mask layer **190** also covers the entire top surface of the second openings **171** (referring to FIG. 5) to protect the initial sidewall layer **181** on the sidewalls of the second openings **171**.

(47) In some embodiments, the material of the second mask layer **190** may be photoresist.

(48) Referring to FIGS. 11 and 12, using the second mask layer **190** (referring to FIG. 10) as a mask, the initial sidewall layer **181** (referring to FIG. 10) located on opposite sidewalls of the first opening **170** (referring to FIG. 5) in the first direction X is removed, and the remaining initial sidewall layer **181** (refer to FIG. 10) serves as the sidewall layers **180**. The second mask layer **190** (referring to FIG. 10) is removed. The initial sidewall layer **181** (referring to FIG. 10) is patterned so that the formed sidewall layers **180** are spaced in the second direction Y, and thus two spaced gates are formed in the subsequent process for forming the gates.

(49) In some embodiments, in the operation of forming the sidewall layer **180**, the sidewall layers **180** located in the second opening **171** (referring to FIG. 5) define a space, that is, the width of the second opening **171** (referring to FIG. 5) is greater than the sum of the thickness of the sidewall layers **180** on both sides during forming the sidewall layer **180**, so that the top surface of the initial isolation layer **140** is also exposed in the second opening **171** (referring to FIG. 5) during forming the sidewall layer **180**.

(50) In some embodiments, in the second direction Y, the length of the second openings **171** (referring to FIG. 5) is larger than the width of the sidewall layer **180** on the sidewall of the first opening **170** (referring to FIG. 5). It could be understood that, if the length of the second openings **171** (referring to FIG. 5) is less than the width of the sidewall layer **180**, in the process of forming the sidewall layer **180** on the sidewalls of the second openings **171** (referring to FIG. 5), since the length of the second opening **171** (referring to FIG. 5) is small, the material of the sidewall layer **180** cannot be totally accommodated along the length of the second opening **171** (referring to FIG. 5), and a part of the material of the sidewall layer **180** may accumulate in a direction away from the activated region **100** during deposition, as a result, the heights of the surface of the sidewall layer **180** are different, and even some unnecessary areas are covered, so that the pattern of the gate subsequently formed is inaccurate. By providing the length of the second opening **171** (referring to FIG. 5) larger than the width of the sidewall layer **180** on the sidewall of the first opening **170** (referring to FIG. 5), the formed sidewall layer **180** can have a better morphology.

(51) In some embodiments, in the first direction X, the width of the formed gap is 0-5 nm. It could

be understood that, the initial gate **110** is patterned with the sidewall layer on the sidewall of the second opening **171** (referring to FIG. 5) as a mask to form a protruding part of the gate. The protruding part is used for contact and connection with the contact structure formed subsequently. When the width of the formed gap is larger, the connection area between the formed contact structure and the protruding part of the gate is smaller; the width of the formed gap is larger, the area of the gate subsequently used for contacting with the contact structure is larger, it is less difficult to form the contact structure on the surface of the gate. When the width of the formed gap is greater than 5 nm, poor contact between the contact structure and the gate may occur.

Consequently, by the width of the formed gap being 0-5 nm, and the area of the gate used for the contacting with the contact structure can be enlarged as much as possible without affecting the contact between the contact structure subsequently formed and the gate.

(52) In some embodiments, the width of the formed second opening **171** (referring to FIG. 5) in the first direction is 8-20 times of the width of the gap. It can be understood that the greater the width of the second opening **171** (referring to FIG. 5), the greater the area of the gate for the contacting with the contact structure, whereas the greater the width of the second opening **171** (referring to FIG. 5), the smaller the area on the activated region **100** for forming other structures. Accordingly, the smaller the width of the second opening **171** (referring to FIG. 5), the smaller the area of the gate for the contacting with the contact structure, whereas the smaller the width of the second opening **171** (referring to FIG. 5), the larger the area on the activated region **100** for forming other structures. By controlling the width of the formed second opening **171** to be 8-20 times of the width of the gap, it is possible to reserve an area for forming other structures on the activated region **100** while increasing the area of the gate for the contacting with the contact structure.

(53) Referring to FIGS. 13 and 14, FIG. 13 and FIG. 14 are operations performed on the basis of FIG. 11 and FIG. 12, and FIG. 14 is a sectional view of FIG. 13 in the direction of the dashed line.

(54) The initial gate **110** is patterned with the sidewall layers **180** on both sides of the first opening **170** (referring to FIG. 5) as a mask to form a gate **200**. In some embodiments, the gate **200** includes a body part and a protruding part on one side of the body part for contacting with a contact structure formed subsequently. In some embodiments, before the initial gate **110** is patterned, the operations for forming the gate **200** include the following operations. The initial intermediate layer **130** and the initial isolation layer **140** are patterned with the sidewall layer **180** as a mask, and the remaining initial intermediate layer **130** serves as the intermediate layer, and the remaining the initial isolation layer **140** is used as the isolation layer; the initial gate **110** is patterned with the intermediate layer, the isolation layer and the sidewall layers **180** together as a mask to form the gate **200**. By using the intermediate layer, the isolation layer and the sidewall layers **180** together as a mask, the pattern accuracy of forming the gate **200** can be improved, thereby improving the stability of the semiconductor structure.

(55) After the gate **200** is formed, the intermediate layer, the isolation layer and the sidewall layers **180** are removed. That is, in the present disclosure, the feature size of the gate **200** is defined by the width of the sidewall layers **180**, so that a smaller feature size of the gate can be obtained and it is more flexible in subsequent circuit design.

(56) An embodiment of the present disclosure also provides another method for manufacturing a semiconductor structure. The manufacturing method is roughly the same as the above-described method, and the main differences is that the mask layer of the semiconductor structure provided in the present disclosure has different patterns, and the structures of the gate formed subsequently are also different. Another method for manufacturing a semiconductor structure provided by the embodiment of the disclosure will be described with reference to the drawings. It should be noted that the same or corresponding parts as the preceding embodiments can refer to the preceding corresponding description, and will not be repeated below.

(57) In this embodiment, taking a positive photoresist as an example, protrusions are formed on the first photoresist and the first mask layer **151** is patterned with the first photoresist layer as a mask.

(58) Referring to FIGS. **15** to **18**, FIG. **15** is a top view provided by the embodiment of the disclosure. FIG. **16** is a sectional view of FIG. **15** along the direction of the dashed line; FIGS. **17** and **18** are operations performed on the basis of FIGS. **15** and **16**, and FIG. **18** is a sectional view of FIG. **17** along the direction of the dashed line.

(59) Specifically, the initial gate **110** is patterned with the first mask layer **151** as a mask, to form two of the gates **200** spaced apart away each other, and the protruding part of one gate **200** faces the other gate **200** located in the same activated region, in other words, the protruding part of one gate **200** is located between the adjacent gates **200**.

(60) It should be noted that, since the protruding part needs to be formed between adjacent gates **200**, it is necessary to satisfy that the width of the first opening **170** in the first direction X is larger than the sum of the length of the protruding part and the width of the sidewall layers **180**, so as to provide a space where the protruding are formed between adjacent gates **200**.

(61) In the method for manufacturing a semiconductor structure provided by the embodiment of the disclosure, two of the gates **200** are formed on the same activated region, thereby providing a process basis for subsequent formation of common source or common drain transistors. In addition, a protruding part is formed at one side of the body part of the gate, thereby providing a larger contact area for the subsequent formation of the contact structure, and thus facilitating the subsequent process, and also reducing the contact resistance between the gates **200** and the contact structure, thereby improving the performance of the semiconductor structure.

(62) Another embodiment of the present disclosure also provides a method for manufacturing a semiconductor structure. The method for manufacturing a semiconductor structure is roughly the same as the above-described embodiments, and the main differences is that the gates are formed on different activated region in another embodiment of the disclosure. The method for manufacturing a semiconductor structure provided by another embodiment of the disclosure will be described with reference to the drawings. It should be noted that the same or corresponding parts as the preceding embodiments can refer to the preceding corresponding description, and will not be repeated below.

(63) In this embodiment of the disclosure, taking a positive photoresist as an example, a protrusion is formed on the first photoresist and the first mask layer is patterned with the first photoresist layer as a mask.

(64) Referring to FIGS. **19** to **22**, FIG. **20** is a sectional view of FIG. **19** along the direction of the dashed line, FIGS. **21** and **22** are operations performed on the basis of FIGS. **19** and **20**, and FIG. **22** is a sectional view of FIG. **21** along the direction of the dashed line.

(65) Specifically, referring to FIGS. **19** and **20**, the activated region **300** may include a substrate **301**, an active area **302** located on the substrate **301**, and isolation structures **303** located on opposite sides of the active area **302**. The isolation structure **303** may include a first isolation structure **304**, a second isolation structure **305**, and a third isolation structure **306**. The initial gate **310** is also provided on the activated region **300**. The initial gate **310** may include a first initial conductive layer **311**, a second initial conductive layer **312** and an initial protective layer **313** which are stacked. A gate dielectric layer **320** is also included between the initial gate **310** and the activated region **300**. An initial intermediate layer **330** and an initial isolation layer **340** are provided, in which the initial intermediate layer **330** is located between the initial isolation layer **340** and the initial gate **310**. A first mask layer **351** is provided, which includes a first opening **370**; an initial sidewall layer **381** is provided at least on a sidewall of the first opening **370**.

(66) In some embodiments, the first mask layer **351** further includes extending parts **371** extending from the first mask layer **351** toward the first opening **370** and connected to opposite sides of the first opening **370** in the second direction Y, and the extending parts **371** are located on different activated regions **300**.

(67) It should be noted that the extending parts are equivalent to the second openings in the above-mentioned embodiment. The first mask layer with the extending parts can provide a process basis for forming the protruding parts of the gates subsequently, thereby increasing the areas of the gates

subsequently for contacting with the contact structure.

(68) In some embodiments, the width of the formed first opening **370** in the second direction Y is greater than the width of the extending part **371** in the first direction X. By controlling the width of the formed first opening **370** to be larger than the width of the extending part **371**, it can be avoided that a part of the material of the initial sidewall layer **381** stacks due to a small space in the process for forming the initial sidewall layer **381**.

(69) Referring to FIGS. **21** and **22**, the sidewall layers **380** are formed by patterning the initial sidewall layer **381** and the sidewall layers **380** are located on different activated regions **300**.

(70) In some embodiments, forming the sidewall layers **380** includes forming the sidewall layers **380** located on the sidewalls of the first opening **370** (referring to FIG. **19**) and the sidewall layers **380** located on the sidewalls of the extending parts **371** (referring to FIG. **19**) in the same step.

(71) In some embodiments, the first opening **370** crosses adjacent two activated regions **300**, and forming the sidewall layers **380** includes forming the sidewall layers **380** above different initial gates **310**. By controlling the first opening **370** to cross two adjacent activated regions **300**, the gates **400** formed subsequently can be located on different activated regions **300** respectively, which provides a process basis for the subsequent formation of transistors.

(72) Referring to FIGS. **23** and **24**, FIG. **23** and FIG. **24** are operations performed on the basis of FIG. **21** and FIG. **22**.

(73) Specifically the initial gate **310** (referring to FIG. **22**) is patterned with the sidewall layers **380** as a mask to form gates located on different activated regions.

(74) Another embodiment of the present disclosure also provides another method for manufacturing a semiconductor structure. The manufacturing method is roughly the same as the above-described method, and the main differences is that the mask layer of the semiconductor structure provided in the present disclosure has different patterns, and the structures of the gate formed subsequently are also different. Another method for manufacturing a semiconductor structure provided by the embodiment of the disclosure will be described with reference to the drawings. It should be noted that the same or corresponding parts as the preceding embodiments can refer to the preceding corresponding description, and will not be repeated below.

(75) In this embodiment, taking a negative photoresist as an example, protrusions are formed on the first photoresist and the first mask layer **151** is patterned with the first photoresist layer as a mask.

(76) Referring to FIGS. **25** to **28**, FIG. **25** is a top view provided by the embodiment of the disclosure. FIG. **26** is a sectional view of FIG. **25** along the direction of the dashed line; FIGS. **27** and **28** are formed on the basis of FIGS. **25** and **26**, and FIG. **28** is a sectional view of FIG. **27** along the direction of the dashed line.

(77) Specifically, the initial gate **310** is patterned with the first mask layer **351** as a mask, to form two spaced gates **400**, in which the protruding part of one of the gates **400** extends away from another gate **400** located on another activated region, in other words, the protruding parts of the gates **400** are located on the outer sidewall of the adjacent gates **400**.

(78) It should be noted that, in the embodiments of the present disclosure, the gates **400** are formed on different activated regions **300**. Usually, there is sufficient space for forming the gates **400**, such that the width of the first opening **370** is not additionally limited in the embodiment of the present disclosure. In other embodiments, the size of the device is reduced to a certain extent, and there may be insufficient space for forming the gates **400**. Therefore, it is necessary to limit the width of the first opening, the length of the protruding parts and the width of the sidewall layers to provide sufficient space for forming the gates.

(79) In the embodiments of the present disclosure, two sidewall layers **380** are formed on different activated regions **300**, and the initial gate **310** is patterned with the sidewall layer **380** as a mask, so that two gates **400** spaced apart away each other can be formed, thereby providing a process basis for forming transistors, and the sidewall layers **380** formed by the first mask layer having the extending parts **371** as a mask also has the same protruding parts, thereby providing more contact

areas for subsequent formation of the contact structure and improving the reliability of the semiconductor structure.

(80) Embodiments of the present disclosure also provide a semiconductor structure which may be formed using part or all of the above-described operations and the same or corresponding parts may refer to the above-described embodiments and will not be repeated below. The semiconductor structure provided by an embodiment of the present disclosure will be described in detail below with reference to the accompanying drawings.

(81) Referring to FIGS. 29 and 30, FIG. 29 is a top view of a semiconductor structure provided by the embodiment of the present disclosure, and FIG. 30 is a sectional view provided by an embodiment of the present disclosure.

(82) Specifically, the semiconductor structure includes an activated region 500; a gate 510 located on the activated region 500, in which the gate 510 includes a body part 511 extending in the first direction X and a protruding part 512 located on one side of the body part 511, the protruding part 512 is connected with a sidewall of the body part 511 extending along the first direction X, and the protruding part 512 and the body part 511 are in an integrated structure; a contact structure 520 located on a surface of the protruding part 512 and electrically connected with the gate 510.

(83) In some embodiments, the activated region 500 may include a substrate 501, an active area 502 on the substrate 501 and isolation structures 503 on opposite sides of the active area 502. The semiconductor structure further includes a gate dielectric layer 550 for avoiding contact between the gate 510 and the activated region 500.

(84) By providing the protruding part 512, the contact area between the gate 510 and the contact structure 520 can be increased, thereby reducing the contact resistance between the contact structure 520 and the gate 510, and improving the performance of the semiconductor structure.

(85) In some embodiments, the body part 511 is also provided with a gap therein, and the gap penetrates the body part 511 in the direction of one protruding part 512 toward the body part 511. It could be understood that the larger the gap, the greater the width of the corresponding protruding part 512 in the first direction X, and the width of the protruding part 512 can be increased by forming the gap, thereby increasing the contact area between the gate 510 and the contact structure 520. In other embodiments, the body part 511 may not have a gap and it may be adjusted according to the actual manufacturing process.

(86) In some embodiments, the top surface of one activated region 500 includes two gates 510, and the protruding parts 512 of the two gates 510 are spaced in the first direction X and in the direction perpendicular to the first direction X. By controlling the protruding parts 512 to be spaced in the first direction X and in the direction perpendicular to the first direction X, the subsequent arrangement of the contact structure 520 can be facilitated and subsequent routing can be facilitated. In other embodiments, the protruding part 512 may also be spaced only in the first direction or in the direction perpendicular to the first direction.

(87) In some embodiments, the semiconductor structure further includes a source 530 and a drain 540, in which the source 530 and the drain 540 are located in the activated region 500, and the same source 530 or the same drain 540 are included between two adjacent ones of the gates 510. The activated region is ion doped to form the source 530 and the drain 540, so that a desired common source transistor or a common drain transistor can be formed as required. The common source transistor or the common drain transistor can be used as part of the sense amplifier.

(88) In some embodiments, a plurality of second contact structures 560 are further included, the second contact structures 560 are spaced on the surface of the activated region 500, and one second contact structure 560 is electrically connected to one source 530 or one drain 540. The signals of the source 530 and drain 540 of the transistor can be derived by providing the second contact structures 560.

(89) The materials of the contact structure 520 and the second contact structures 560 may be the same, for example both may be metal.

(90) In the embodiments of the disclosure, by providing the gate **510** having the protruding part **512**, the contact area between the contact structure **520** and the gate **510** is increased, thereby reducing the contact resistance between the contact structure **520** and the gate **510**, and improving the performance of the semiconductor structure.

(91) Those of ordinary skill in the art will appreciate that the above-described embodiments are specific embodiments for implementing the present disclosure and that in practical application various changes may be made in form and detail thereof without departing from the spirit and scope of the embodiments of the present disclosure. Any person skilled in the art may make their own changes and modifications without departing from the spirit and scope of the embodiments of the present disclosure, so the protection scope of the embodiments of the present disclosure shall be subject to the scope defined by the claims.

Claims

1. A method for manufacturing a semiconductor structure, comprising: providing an activated region; forming an initial gate located on the activated region; forming a first mask layer on a top surface of the initial gate, wherein a first opening penetrating the first mask layer is provided in the first mask layer, and the first opening at least has opposite two sides extending along a first direction, the first mask layer further comprises second openings, the second openings are located on opposite two sides of the first opening along a second direction and are communicated with the first opening; forming sidewall layers located on sidewalls of the two sides of the first opening extending along the first direction and the sidewall layers located on sidewalls of the second openings in a same process; removing the first mask layer; patterning the initial gate to form gates, each gate comprises a body part and a protruding part, wherein the body part is formed by the sidewall layers of the two sides of the first opening as a mask, and the protruding part is formed with the sidewall layers of the sidewalls of the second openings as a mask.
2. The method for manufacturing a semiconductor structure according to claim 1, wherein during forming the sidewall layers, the sidewall layer in one second opening defines a gap.
3. The method for manufacturing a semiconductor structure according to claim 1, wherein a length of the second openings is larger than a width of the sidewall layers on sidewalls of the first opening in the second direction.
4. The method for manufacturing a semiconductor structure according to claim 1, wherein a length of the second openings is larger than a width of the sidewall layers on sidewalls of the first opening in the second direction.
5. The method for manufacturing a semiconductor structure according to claim 4, wherein a width of the gap formed in the first direction is 0 to 5 nm.
6. The method for manufacturing a semiconductor structure according to claim 2, wherein a width of the second opening formed in the first direction is 8-20 times of a width of the gap.
7. The method for manufacturing a semiconductor structure according to claim 1, wherein forming the sidewall layers comprises: forming an initial sidewall layer located on the sidewalls of the first opening and a bottom of the first opening, a top surface of the initial sidewall layer in the first opening being lower than a top surface of the first mask layer; and removing the initial sidewall layer located at a part of the bottom of the first opening, and removing the initial sidewall layer located on sidewalls of the first opening which are opposite in the first direction to form the initial sidewall layer spaced in a direction perpendicular to the first direction, the remaining initial sidewall layer serving as the sidewall layers.
8. The method for manufacturing a semiconductor structure according to claim 7, wherein during forming the initial sidewall layer, the initial sidewall layer is also formed on a top surface of the first mask layer; a maskless etching process is adopted to simultaneously remove the initial sidewall layer on the part of the bottom of the first opening and on the top surface of the first mask

layer.

9. The method for manufacturing a semiconductor structure according to claim 7, wherein removing the initial sidewall layer located on sidewalls of the first opening which are opposite in the first direction comprises: forming a second mask layer after removing the first mask layer, the second mask layer exposing the initial sidewall layer on the sidewalls of the first opening along the first direction; removing the initial sidewall layer on the sidewalls of the first opening which are opposite in the first direction with the second mask layer as a mask; and removing the second mask layer.

10. The method for manufacturing a semiconductor structure according to claim 1, wherein the first opening crosses two adjacent ones of the activated regions, and forming the sidewall layers comprises forming the sidewall layers above different initial gates.

11. The method for manufacturing a semiconductor structure according to claim 10, wherein the first mask layer further comprises extending parts, wherein the extending parts extend from the first mask layer toward the first opening and are connected with the two sides of the first opening which are opposite in a second direction, and the extending parts are located on different activated regions, and wherein forming the sidewall layers comprises: forming the sidewall layers located on the sidewalls of the first opening and the sidewall layers located on sidewalls of the extending parts in a same process.

12. The method for manufacturing a semiconductor structure according to claim 11, wherein a width of the first opening in the second direction is greater than a width of the extending parts in the first direction.

13. The method for manufacturing the semiconductor structure according to claim 1, wherein, forming the gates comprises: forming an initial intermediate layer and an initial isolation layer on a surface of the initial gate, wherein the initial intermediate layer is located between the initial isolation layer and the initial gate; and wherein before patterning the initial gate, the method further comprises: patterning the initial intermediate layer and the initial isolation layer with the sidewall layers as a mask, the remaining initial intermediate layer serving as an intermediate layer and the remaining initial isolation layer serving as an isolation layer; and patterning the initial gate with the intermediate layer, the isolation layer and the sidewall layers together as a mask to form the gates.
